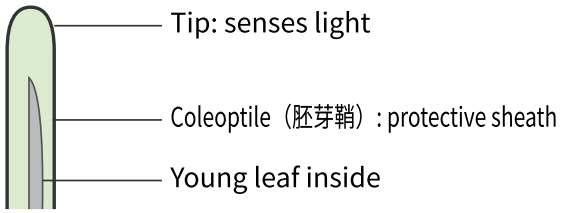
Name:_____ Class:_____ Student number:_____

Light sensing and growth in plants

Read A, then complete B. Read C only after finishing B. You do not need to operate a virtual lab.

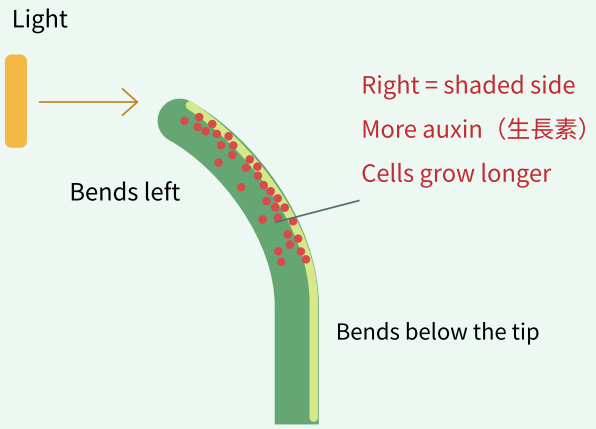


An oat coleoptile（胚芽鞘） is a sheath that surrounds and protects the young leaf. This activity examines elongation（延長） and bending of the sheath.

Learn the core concepts first

1. Directional plant growth in response to light direction is called **"phototropism（向光性）"**; growth towards light is called "positive phototropism（向光性）".
2. In this oat coleoptile（胚芽鞘） model, **Tip** helps **sense light from one side**.
3. Other research helps us understand the substance called "substance X" in this inquiry as **Auxin（生長素）**. Auxin（生長素） is a hormone that affects plant growth; the tip can produce a growth-promoting effect that **passes downwards**.
4. With light from one side, **more auxin（生長素） is found on the shaded side, so cells on that side elongate（延長） more, making the coleoptile（胚芽鞘） bend towards light**.

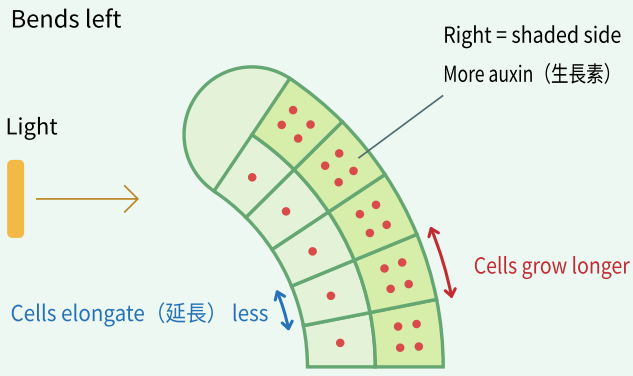
1 • Light from the left



Tip senses light → growth signal passes down

- Red dots = auxin（生長素）

2 • Light from the left: enlarged cells



More elongation（延長） on the shaded side → bends left

Worked example: a reasoned prediction

Given prediction: ‘The coleoptile（胚芽鞘） may still bend towards light when light is blocked below the tip.’ This is reasonable: the intact tip can sense light from one side and a growth-promoting effect can pass down. The lower region need not receive light directly to elongate（延長） unequally. Compare A and C to test it.

Example: identify variables

A–C changes only the shading treatment and records bending. Keep light direction and intensity, species, initial size and condition, time, temperature and water the same. The independent variable is deliberately changed; the dependent variable is the response observed or measured; controlled variables are kept the same.

Tip removal and agar（瓊脂）: apply the concepts to new comparisons

The tip senses light and can produce an effect that promotes elongation（延長） below it. A prediction that tip removal cannot affect elongation（延長） overlooks this role; check the A/D data. Cutting also creates a wound and removes tissue, so this comparison alone cannot prove a unique mechanism.

Blank agar（瓊脂） has not touched the tip; treated agar（瓊脂） has touched the tip. While agar（瓊脂） is in contact with a coleoptile（胚芽鞘） tip, a growth-promoting substance can diffuse from the tip into the agar（瓊脂）. Later, when this agar（瓊脂） is placed on a coleoptile（胚芽鞘） with its tip removed, the growth-promoting substance can diffuse out of the agar（瓊脂） and into the coleoptile（胚芽鞘）. Comparisons of treatment and placement occur in darkness; bending cannot be attributed to light acting during incubation.



Method example only: 20°, right

Worked example: angle and direction are separate

Use the vertical line as 0°. Read how far the tip direction differs from vertical. This example is 20° to the right. The same tilt to the left is still 20°, but its direction is left.

The solid pointer follows the tip direction; the dashed line is the vertical reference. This diagram teaches reading, and is not experimental data for E–H.

The 24-hour period and angles are teaching-model settings, not real constants. Elongation（延長） and bending do not directly measure auxin（生長素） concentration or identify a chemical substance. Auxin（生長素） identity and its distribution under light are understood together with other research.